

## ELECTRODYNAMICS PROBLEMS

**Q1. Convert the point P(1,3,5) from Cartesian to Cylindrical and Spherical polar coordinates**

**Given:-** X= 1; Y= 3; Z=5

**Formula:-** Cylindrical- Cartesian  $r = \sqrt{x^2 + y^2}$ ;  $\phi = \tan^{-1} \left( \frac{y}{x} \right)$ ;  $z = z$

Spherical-Cartesian;

$$r = \sqrt{x^2 + y^2 + z^2}; \theta = \tan^{-1} \left( \frac{\sqrt{x^2 + y^2}}{z} \right); \phi = \tan^{-1} \left( \frac{y}{x} \right)$$

**Solution:-**

In cylindrical polar coordinates,  $(r, \phi, z)$

$$r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 3^2} = \sqrt{10} = 3.162$$

$$\phi = \tan^{-1} \left( \frac{y}{x} \right) = \tan^{-1}(3) = 71.57^\circ$$

$$Z = 5$$

In Spherical polar coordinates  $(r, \theta, \phi)$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 3^2 + 5^2} = \sqrt{35} = 5.916$$

$$\phi = \tan^{-1} \left( \frac{y}{x} \right) = \tan^{-1}(3) = 71.57^\circ$$

$$\theta = \tan^{-1} \left( \frac{\sqrt{x^2 + y^2}}{z} \right) = \tan^{-1} \left( \frac{\sqrt{1^2 + 3^2}}{5} \right) = 32.3^\circ$$

**Ans:- Coordinates of P in cylindrical coordinates are (3.162, 71.57, 5) and in spherical polar coordinates are (5.916, 71.57, 32.3)**

**Q2.** Given vector  $\vec{A}(x, y, z) = y\hat{i} + (x + z)\hat{j}$  in Cartesian coordinate system at point P(-2,6,3). Convert the vector  $\vec{A}$  into cylindrical and spherical coordinates.

**Given:-**  $\vec{A} = y\hat{i} + (x + z)\hat{j}$ ,  $x=-2, y=6, z=3$

**Formula:-** Cylindrical- Cartesian  $r = \sqrt{x^2 + y^2}$ ;  $\phi = \tan^{-1} \left( \frac{y}{x} \right)$ ;  $z = z$

Spherical-Cartesian;

$$r = \sqrt{x^2 + y^2 + z^2}; \theta = \tan^{-1} \left( \frac{\sqrt{x^2 + y^2}}{z} \right); \phi = \tan^{-1} \left( \frac{y}{x} \right)$$

**Solution:-**

Cylindrical coordinates:

$$r = \sqrt{(-2)^2 + 6^2} = 6.32$$

$$\phi = \frac{6}{-2} = 108.43^\circ$$

$$z=3$$

Spherical coordinates:

$$r = \sqrt{(-2)^2 + 6^2 + 3^2} = 7$$

$$\theta = \frac{\sqrt{(-2)^2 + 6^2}}{3} = 64.62^\circ$$

$$\phi = \frac{6}{-2} = 108.43^\circ$$

**Ans:-**  $\vec{A} = (6.32 \hat{r}, 108.43^\circ \hat{\phi}, 3 \hat{z})$  is cylindrical coordinates

**$\vec{A} = (7 \hat{r}, 64.62^\circ \hat{\theta}, 108.43^\circ \hat{\phi})$  is spherical coordinates.**

**Q3. Find the divergence and curl of the field  $F = 30\hat{i} + 2xy\hat{j} + 5xz^2\hat{k}$  in Cartesian coordinates.**

**Given:-**  $F = 30\hat{i} + 2xy\hat{j} + 5xz^2\hat{k}$

**Formula:-** Divergence  $(\vec{\nabla} \cdot \vec{A}) = \hat{i} \frac{\partial}{\partial x} A_x + \hat{j} \frac{\partial}{\partial y} A_y + \hat{k} \frac{\partial}{\partial z} A_z$

$$\text{Curl: } (\vec{\nabla} \times \vec{A}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

**Solution:-**

$$\begin{aligned} \text{Divergence, } \vec{\nabla} \cdot \vec{F} &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (30\hat{i} + 2xy\hat{j} + 5xz^2\hat{k}) \\ &= \frac{\partial}{\partial x} (30) + \frac{\partial}{\partial y} (2xy) + \frac{\partial}{\partial z} (5xz^2) = 2x + 10xz \\ \vec{\nabla} \cdot \vec{F} &= 2x(1 + 5z) \end{aligned}$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 30 & 2xy & 5xz^2 \end{vmatrix} = -5z^2 \hat{j} + 2y \hat{k}$$

**Ans:- Divergence of field is  $2x(1 + 5z)$  and its curl  $= -5z^2 \hat{j} + 2y \hat{k}$**

**Q4. If  $\phi(x, y, z) = 3(x^2y - y^2x)$ , Calculate gradient.**

**Given:-**  $\phi(x, y, z) = 3(x^2y - y^2x)$

**Formula:-**

**Solution:-**

$$\begin{aligned} \text{Grad } \phi &= \nabla \phi = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \{3(x^2y - y^2x)\} \\ &= \hat{i}(6xy - 3y^2) + \hat{j}(3x^2 - 2yx) + \hat{k}(0) \\ \text{Therefore, } \nabla \phi &= \hat{i}(6xy - 3y^2) + \hat{j}(3x^2 - 2yx) \end{aligned}$$

**Ans:- The gradient is ,  $\nabla \phi = \hat{i}(6xy - 3y^2) + \hat{j}(3x^2 - 2yx)$ .**

**Q5.** A region is specified by the potential function given by  $\phi = 4x^2 + 3y^2 - 9z^2$ . Calculate electric field strength.

**Given:-**  $\phi = 4x^2 + 3y^2 - 9z^2$ .

**Formula:-**  $\vec{E} = -\text{grad}(\text{potential function})$

**Solution:-**

$$\begin{aligned}\vec{E} &= -\text{grad}(\text{potential function}) \\ &= -\text{grad } \phi = -\nabla[4x^2 + 3y^2 - 9z^2] \\ &= -\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right)(4x^2 + 3y^2 - 9z^2)\end{aligned}$$

Therefore,  $\vec{E} = -8x\hat{i} - 6y\hat{j} + 18z\hat{k}$

**Ans:- The electric field strength is  $\vec{E} = -8x\hat{i} - 6y\hat{j} + 18z\hat{k}$ .**